

Raaga Suravarapu

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Computer Science undergraduate with hands-on experience in data science, machine learning, and NLP research, including model benchmarking, analytics dashboards, and production ML systems. Strong ability to analyze data, communicate insights to technical and non-technical stakeholders, and apply responsible AI principles to drive equitable, real-world impact in financial services and technology.

Education

Virginia Tech, Blacksburg, VA

B.S. in Computer Science (General Engineering) — Expected May 2029

Oakridge International School, Hyderabad, India

International Baccalaureate Diploma Programme — Graduated May 2025

Honor Roll IB MYP e-assessments (52/56, Grade 10), Honor Roll (April 2024) – Spanish ab Initio DP1

Experiences

NLP Undergraduate Research Fellow w/ Dr. Gray, Virginia Tech, Blacksburg, VA | Aug 2025 – Present

- Benchmarked **5+ transformer-based sentence embedding** models, improving **semantic similarity** relevance by ~12% across evaluation datasets
- Optimized **sentiment classification** performance through **prompt engineering** and **model configuration** experiments
- Built interpretable **visual analytics dashboards** (aspect coverage, keyword frequency) to tune filtering thresholds and support model decisions
- Integrated end-to-end **NLP pipelines** with a front-end dashboard, enabling all users to explore and validate insights
- Evaluated bias and fairness metrics to identify and **mitigate skewed outcomes** across demographic and thematic groups

IT/Prompt Engineering Intern BeyondScale, Hyderabad, Telangana, India | May 2023 - July 2023

- Contributed to an AI-driven **personalized learning platform** used by live customers, supporting real-time recommendation workflows
- Improved machine learning model accuracy by ~15% through **Python-based data cleaning, feature selection, and preprocessing**
- Analyzed structured and semi-structured datasets using **Pandas and NumPy** to identify **patterns** impacting model performance
- Collaborated with engineers to **test, debug, and deploy ML updates**, gaining exposure to production-level AI systems

Software Development Intern, Amrutha Hastham Charitable Trust, Hyderabad, India | May 2024 - June 2024

- Engineered secure donation processing by **integrating Razorpay API** into a mobile app, enabling \$30000+ in transactions with full **error handling** and **verification**
- Authored **technical documentation** and deployment guidelines for future maintainers
- Worked **cross-functionally** with design and product leads to ensure a smooth user experience

Chief Human Resources Officer (CHRO) - Marise Foundation | Jul 2022 – Sep 2024

- Led **recruitment, onboarding, and coordination** for a 40+ member volunteer organization
- Streamlined onboarding workflows, improving **volunteer retention** and **operational efficiency**
- Maintained structured records and **supported leadership** with **data-backed insights** on team capacity and **engagement**

Projects

Research paper on “Natural Language AI Prototype: Retrieval Augmented Generation” | 2024

- Built AI prototype delivering personalized educational content using **RAG architecture**.
- Benchmarked performance and scalability and achieved **92% query relevance** in testing.

Housing Price Prediction, IB Extended Essay | 2023

- Conducted a **comparative study** of Random Forest vs. Multiple Linear Regression models
- Preprocessed large **datasets** and evaluated performance using **RMSE and R² metrics**
- Demonstrated strengths of **ensemble models** for complex, real-world data

Skills

Python, Java, SQL, JavaScript (React), Machine Learning, Statistical Analysis, Feature Engineering, Model Evaluation, Pandas, NumPy, Scikit-learn, Git, Figma, REST APIs, Agile & project-based work, documentation, cross-functional work, commitment to diversity, inclusion, and community impact